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Navigate: an open-source platform for smart light-sheet microscopy

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Light-sheet fluorescence microscopy (LSFM) enables fast, gentle and large-volume imaging, making it an increasingly important tool for cell biological discovery. In both live cells and fixed tissues, many important cellular interactions and microenvironmental niches are exceedingly rare. Automatic identification of such events is computationally expensive, especially if such analyses are performed for many biological replicates or through time. In living specimens, such as developing embryos, cellular dynamics often require an experimenter to not only detect the initial event, but manually update the microscope's configuration to track movement or risk losing the sample. To address such challenges, researchers have started to apply smart microscopy techniques to LSFM[1–3]. Despite smart imaging approaches existing for over a decade[4], most LSFMs operate in a stubbornly classical operation paradigm. In part, this is because each LSFM is optimized for specific biological contexts with distinct hardware and control requirements, which may not be supported by existing smart routines. Compounding the problem, most LSFM control software has been built in a piecemeal fashion by combining closed-source LabView or Java code with separate Java or Python routines that must be installed and/or compiled separately (Supplementary Table 1), making it difficult for users without advanced programming capabilities to adapt and reuse available smart routines. To address these challenges and democratize smart LSFM, we developed navigate, an open-source, Python-based LSFM control software with easily reconfigured hardware controls and graphical user interface (GUI)-reconfigurable smart acquisition routines (Fig. 1a,b).

Navigate enables biologists and technology developers alike to establish and reuse smart microscopy pipelines on diverse sets of hardware from within a single framework. While generalizable Python-based frameworks for smart microscopy have been built, they were designed for stimulated emission depletion[5] or single-molecule localization microscopy[6] and do not yet address LSFM's specific acquisition challenges, including decoupled illumination and detection optomechanics and a lack of an optical substrate for focus maintenance. While GUI-based frameworks for image post-processing exist[6], to the authors' knowledge, navigate is the only software that enables decision-based acquisition routines to be generated in a code-free format (Supplementary Table 1).

A schematic of navigate's software architecture is presented in Supplementary Fig. 1. The plug-in architecture of navigate facilitates the addition of new hardware, enabling users to integrate otherwise unsupported devices. For image-based feedback, custom analysis routines can also be loaded within navigate's environment to evaluate images stored as NumPy arrays in memory. Navigate supports the addition of REST-API interfaces for two-way communication with image analysis programs running outside of Python or in different Python environments, such as Ilastik[7], enabling developers to make calls to state-of-the-art software while avoiding dependency conflicts. Image-based feedback can be leveraged to perform diverse tasks, such as sensorless adaptive optics in optically complex specimens (Fig. 1c). We believe this flexibility is necessary for the software to accommodate the diverse modalities of LSFM and to integrate feedback mechanisms.

Navigate can readily be installed and operated without knowledge of software development or an expertise in light-sheet microscopy. Out of the box, it is equipped with an intuitive and rigorously tested GUI (Supplementary Fig. 2) and can capture rich biological data in both classical (Fig. 1d) and smart (Fig. 1e) formats from axially swept, digitally scanned, oblique plane, field synthesis, projection light-sheet imaging modalities in both laser- and sample-scanning formats. Navigate virtualizes microscope control, allowing multiple microscopes to be operated on the fly using a single set of hardware. Navigate writes to next-generation, pyramidal BigDataViewer H5/N5 and OME-Zarr data formats (Supplementary Note 1 and Supplementary Table 2)[8,9], enabling easy movement from acquisition into established image registration and analysis pipelines and fast visual feedback in ImageJ. Data generated adheres to Open Microscopy Environment standards and imports readily into OMERO image data repositories. To ensure academic reproducibility, consistency and reliability, navigate is well documented, and each release of navigate is versioned, archived and given a citable digital object identifier. Because navigate is open-source, is organized in a model–view–controller architecture and includes a plug-in interface, it readily integrates third-party packages such as pymmcore, acquire and more (Supplementary Note 2). We believe its extensibility, user-friendliness even for non-experts, and reproducibility position navigate as the turnkey solution for bespoke, smart LSFM systems.

While initially designed for LSFM, navigate's versatile framework makes it suitable for a broad range of camera-based imaging modalities. Thus, we hope navigate empowers users from all scientific backgrounds to construct and implement novel intelligent imaging workflows, greatly improving their ability to unravel the cellular complexities of life.

Data availability

Imaging data presented in figures are available upon request.

Code availability

The source code for navigate is available on GitHub (https://github.com/TheDeanLab/navigate) and PyPI (https://pypi.org/project/navigate-micro/). Queries and feedback can be provided by creating an issue on GitHub. Documentation on how to extend navigate is provided in the Supplementary Documentation (and is kept up to date at https://thedeanlab.github.io/navigate/). A plug-in template is available at https://github.com/TheDeanLab/navigate-plugin-template.

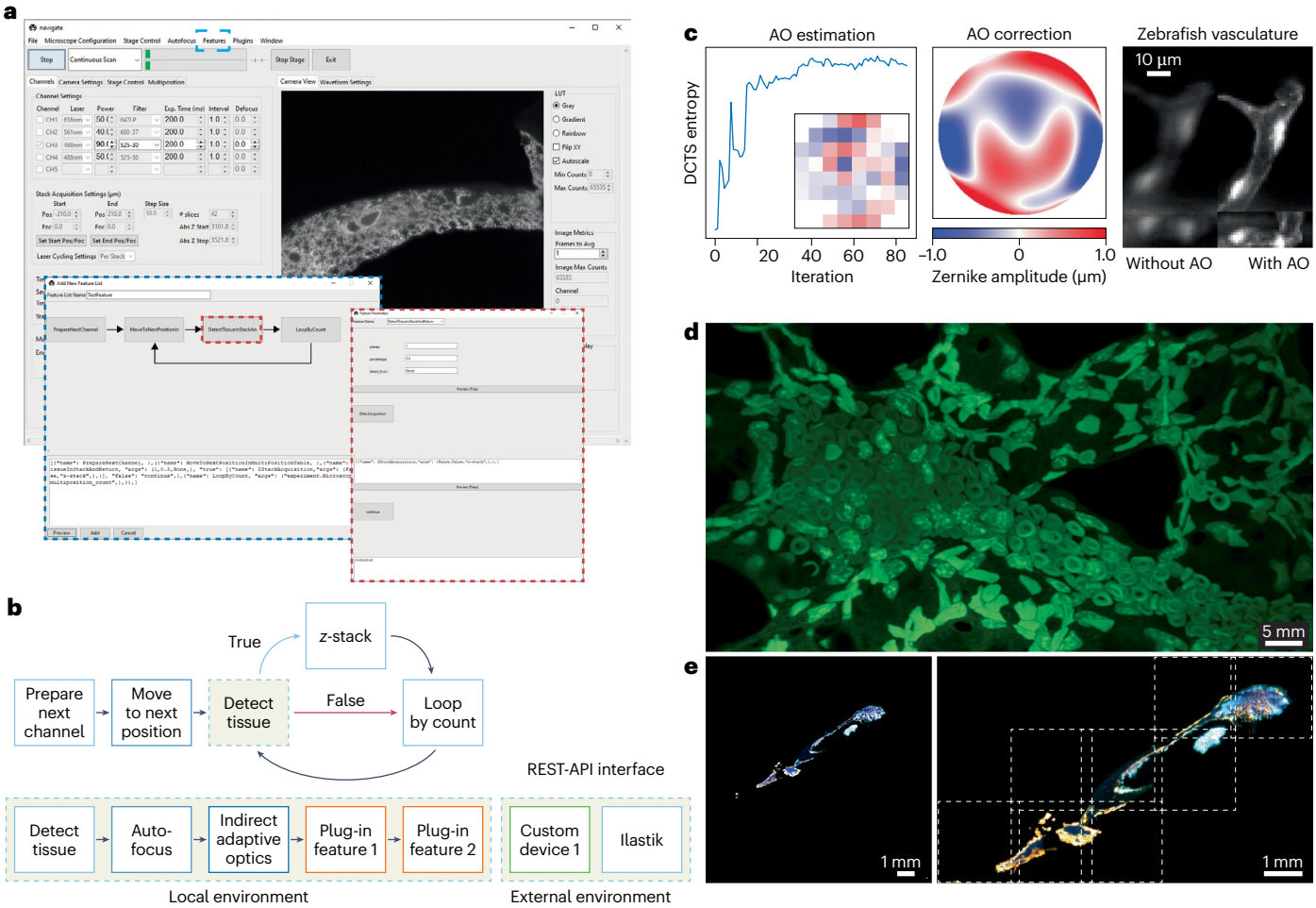

Fig. 1 | Navigate provides a GUI with access to reconfigurable smart acquisition pipelines and runs on multiple types of light-sheet configurations. a, Screenshot of the navigate GUI running a mesoscale selective plane-illumination microscope (mesoSPIM)[10] with GUI controls for the smart acquisition routine diagrammed in **b**. The blue dashed boxes show the Feature menu, where the controls are accessed, and the corresponding editor for the smart routine. The red dashed boxes show a decision node and the corresponding editor for this decision node. **b**, An example of a reconfigurable, decision-based pipeline that acquires a *z*-stack if tissue is present at a particular stage position. The Supplementary User Guide features a tutorial on how to build this pipeline. Individual features can be swapped for other features that ship with the software or with custom-built plug-in features. **c**, Navigate can integrate advanced optical techniques, such as adaptive optics. Left, an iterative sensorless adaptive optics scheme uses the discrete cosine transform Shannon (DCTS) entropy to estimate aberrations. Inset shows the final actuator positions of the deformable mirror. Middle, final estimated adaptive optics wavefront. Right, 3D stack of a ventricle within the vasculature of a zebrafish embryo (labeled with kdrl:eGFP), before and after adaptive optics. **d**, Example image of an expanded mouse lung captured with an upright cleared-tissue axially swept light-sheet microscope (ctASLM) using navigate. **e**, Results of smart imaging workflow from **a** and **b**, captured on a mesoSPIM. Bone tissue imaged at 1× magnification (left) was automatically segmented and tiled at 6× magnification (right). Imaged regions, indicated by dashed white boxes, were saved to a BigDataViewer file format and displayed in a tiled format.

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Author contributions

Z.M., X.W., D.W.C. and K.M.D. designed the software. Z.M., X.W., D.W.C., B.C., C.M., D.M., S.D., S.J.S., P.T., D.P.S. and K.M.D. wrote the code. Z.M., J.L., B.C., C.M., H.M.B. and K.M.D. tested the software. Q.S., H.M.B., S.G., B.A.P. and S.D.C. provided biological samples. C.M. and R.F. implemented adaptive optics routines. K.M.D. supervised the design and development of software, microscopes and tissue preparation methods. Z.M., X.W. and K.M.D. wrote the manuscript. All authors discussed the project and edited the manuscript.

Competing interests

K.M.D. and R.F. have a patent covering ASLM (US10989661) and consultancy agreements with 3i, Inc (Denver, CO, USA). K.M.D has an ownership interest in Discovery Imaging Systems, LLC. The remaining authors declare no competing interests.

Additional information

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